MA5415 QUANTITATIVE RISK MANAGEMENT

Lecture-Exact Formula Cheat Sheet & Solved Exam Problems (Chapters I–V)

Prepared by Barış Arslan — M.Sc. Mathematics Candidate at TU Munich (TUM)

Chapter I and II: Generalized Inverses, Monetary/Coherent Risk Measures, VaR and AVaR

- **Generalized Inverse (Def. 1 & Prop. 1):** $h^{\leftarrow}(y) := \inf\{x \in \mathbb{R} : h(x) \geq y\}$. If h is non-decreasing and right-continuous: $h(x) \geq y \iff x \geq h^{\leftarrow}(y)$ and $h(x) < y \iff x < h^{\leftarrow}(y)$.
- **Lemma 3 (Monotone Transformation):** For u increasing and left-continuous, $F^{\leftarrow}_{u(X)}(y) = u \circ F^{\leftarrow}_X(y)$.
- **Risk Measure Axioms on Discounted P&L $X \in \mathcal{X}$ (Def. 4):** (1) *Monotonicity:* $X \leq Y \implies \rho(X) \geq \rho(Y)$; (2) *Cash-Invariance:* $\rho(X+m) = \rho(X) - m$; (3) *Convexity:* $\rho(\lambda X + (1-\lambda)Y) \leq \lambda\rho(X) + (1-\lambda)\rho(Y)$; (4) *Positive Homogeneity:* $\rho(\lambda X) = \lambda\rho(X)$ $(\lambda \geq 0)$; (5) *Subadditivity:* $\rho(X+Y) \leq \rho(X) + \rho(Y)$.
- **Value-at-Risk & Average Value-at-Risk at level $\lambda \in (0,1)$ (Def. 6 & 8):**

$$\mathrm{VaR}_\lambda(X) := \inf\{m \in \mathbb{R} : \mathbb{P}(m + X < 0) \leq \lambda\} = F^{\leftarrow}_{-X}(1-\lambda), \qquad \mathrm{AVaR}_\lambda(X) := \frac{1}{\lambda}\int_0^\lambda \mathrm{VaR}_\gamma(X)\, d\gamma.$$

- **Normal Position $X \sim \mathcal{N}(\mu, \sigma^2)$:** $\mathrm{VaR}_\lambda(X) = -\mu + \sigma\Phi^{-1}(1-\lambda)$, $\mathrm{AVaR}_\lambda(X) = -\mu + \frac{\sigma}{\lambda}\phi(\Phi^{-1}(1-\lambda))$.
- **Student-t_ν Position $Y \sim t_\nu$:** $\mathrm{VaR}_\lambda(Y) = t^{\leftarrow}_\nu(1-\lambda)$, $\mathrm{AVaR}_\lambda(Y) = \left(\frac{\nu + (t^{\leftarrow}_\nu(1-\lambda))^2}{\nu - 1}\right)\frac{1}{\lambda} f_\nu(t^{\leftarrow}_\nu(1-\lambda))$.

Chapter IV: Extreme Value Theory (MDA, Gnedenko Theorem, POT and Hill Estimator)

- **GEV Distribution $G_\gamma(x) = \exp\{-(1+\gamma x)^{-1/\gamma}_+\}$ & Gnedenko's Theorem (Lemma 8):** With $U(t) = F^{\leftarrow}(1 - 1/t)$:
  1. **Fréchet ($\gamma > 0$):** $F \in \mathrm{MDA}(G_\gamma) \iff x_F = \infty$ and $\overline{F} \in \mathcal{R}_{-1/\gamma}$. Norming: $\tilde{a}_n = U(n), \tilde{b}_n = 0 \implies \Phi_{1/\gamma}(x) = e^{-x^{-1/\gamma}}$.
  2. **Weibull ($\gamma < 0$):** $F \in \mathrm{MDA}(G_\gamma) \iff x_F < \infty$ and $\overline{F}(x_F - x^{-1}) \in \mathcal{R}_{1/\gamma}$. Norming: $\tilde{a}_n = x_F - U(n), \tilde{b}_n = x_F \implies \Psi_{-1/\gamma}(x) = e^{-(-x)^{-1/\gamma}}$ $(x < 0)$.
- **Hill Estimator ($\gamma > 0$, Slide 90):** $\hat{\gamma}_{H_{n,k}} = \frac{1}{k}\sum_{i=0}^{k-1}(\ln X_{n-i,n} - \ln X_{n-k,n})$, $\widehat{\mathrm{VaR}}_{1-\lambda}(X) = X_{n-k,n}\left(\frac{k}{\lambda n}\right)^{\hat{\gamma}_{H_{n,k}}}$.
- **Mean Excess of GPD $H_{\gamma,\sigma}(x) = 1 - (1+\gamma x/\sigma)^{-1/\gamma}$:** For $\gamma < 1$, $e(t) = \mathbb{E}[X - t \mid X > t] = \frac{\sigma + \gamma t}{1-\gamma}$.

Chapter V: Copulas, Concordance, Tail Dependence and Archimedean / EV Copulas

- **Sklar's Theorem & Survival Copula:** $F(x_1, x_2) = C(F_1(x_1), F_2(x_2))$ and $\overline{F}(x_1, x_2) = \hat{C}(\overline{F}_1(x_1), \overline{F}_2(x_2))$, where $\hat{C}(u_1, u_2) = u_1 + u_2 - 1 + C(1-u_1, 1-u_2)$. Fréchet-Hoeffding: $W_d(\mathbf{u}) \leq C(\mathbf{u}) \leq M_d(\mathbf{u})$.
- **Kendall's τ & Spearman's ρ_S:** $\tau_C = 4\mathbb{E}[C(U_1, U_2)] - 1$, $\rho_{S,C} = 12\int_0^1\int_0^1 C(u_1, u_2)\, du_1 du_2 - 3$. For Gaussian/$t_\nu$: $\tau = \frac{2}{\pi}\arcsin(\rho)$, $\rho_S = \frac{6}{\pi}\arcsin(\rho/2)$.
- **Archimedean Copulas $C_\varphi(\mathbf{u}) = \varphi(\varphi^{-1}(u_1) + \cdots + \varphi^{-1}(u_d))$:** $\tau = 1 - 4\int_0^\infty x(\varphi^{(1)}(x))^2 dx$, $\mathrm{LTDC} = 2\lim_{x\to\infty}\frac{\varphi^{(1)}(2x)}{\varphi^{(1)}(x)}$, $\mathrm{UTDC} = 2 - 2\lim_{x\searrow 0}\frac{\varphi^{(1)}(2x)}{\varphi^{(1)}(x)}$.

Sample Exam Problem: Piece-wise Loss Distribution VaR and AVaR Calculation

Question: Let a financial position X have loss distribution $F_{-X}(x) = 0$ for $x < 2$, $F_{-X}(x) = 1 - \frac{2}{x+2}$ for $x \in [2, 6)$, and $F_{-X}(x) = 1 - \frac{9}{x^2}$ for $x \geq 6$. Compute $\mathrm{VaR}_\lambda(X)$ and $\mathrm{AVaR}_\lambda(X)$ for $\lambda \in (0,1)$.
Solution: Note the jump at $x = 2$ where $F_{-X}(2) = 0.5$, and continuity at $x = 6$ where $F_{-X}(6) = 0.75$. Inverting $F_{-X}(x) \geq 1 - \lambda$ gives $\mathrm{VaR}_\lambda(X) = 3/\sqrt{\lambda}$ for $\lambda \in (0, 0.25)$; $\mathrm{VaR}_\lambda(X) = \frac{2}{\lambda} - 2$ for $\lambda \in [0.25, 0.5)$; and $\mathrm{VaR}_\lambda(X) = 2$ for $\lambda \in [0.5, 1)$. Integrating $\frac{1}{\lambda}\int_0^\lambda \mathrm{VaR}_\gamma(X)\, d\gamma$ across the three regimes yields $\mathrm{AVaR}_\lambda(X) = 6/\sqrt{\lambda}$ on $(0, 0.25)$; $\frac{3.5 + 2\ln(4\lambda)}{\lambda} - 2$ on $[0.25, 0.5)$; and $\frac{1.5 + 2\ln 2}{\lambda} + 2$ on $[0.5, 1)$.

PREPARING FOR THE MA5415 QRM EXAM? GET THE COMPLETE SURVIVAL KIT!

This free summary is a preview of the **Complete MA5415 QRM Exam Survival Kit (7 Chapters, 2 Full Mock Exams)**.
The Full Premium Edition includes:

- **All 5 Lecture Chapters Summarized** in exact TUM notation ($F^{\leftarrow}$, VaR_λ, AVaR_λ, $\mathrm{MDA}(G_\gamma)$, $\hat{\gamma}_{H_{n,k}}$, $\hat{C}$, $A(x)$).
- **2 Full-Length 40-Point Timed Mock Exams** (Standard & Advanced Exam Difficulty) with step-by-step proofs.
- **R Simulation Templates** for PIT, QQ-Plots, Hill Estimation, and Marshall-Olkin Copula Simulation.

→ CLICK HERE TO DOWNLOAD THE FULL SURVIVAL KIT ON GUMROAD ←